

Please fill in your answers in the following, also inserting figures/tables if requested. Keep the original questions in the document. Once complete, convert it in a .pdf file and rename it surname_matriculation_01052022_bulk.pdf .

Fill in the following:

Surname: De Benedictis

Matriculation Number: 967528

Master Degree: Bioinformatics for computational genomics

Datasets (fill with one row per tissue)

Liver 5,6,7

Brain 2,3,4

Colon 8,9,10

Tissue Tissue number Column replicate Column replicate Column replicate 123

Your data are in the “RangedSummarizedExperiment” format. They were produced by the GTEx project. The columns contain a lot of information, that permit to look up each sample in the GTEx database for further details, e.g. age/sex of the donor, a picture of the tissue sample itself, and so on.

Question 1:

The counts in each sample have already been “scaled” to a library size of 40M reads (the actual count in each sample can be lower for reads not mapping on exons of annotated genes).

After filtering out from each tissue/replicate table 1) genes shorter than 200 bp and 2) genes annotated on the mitochondrial DNA, fill in the following table, adding one separate row for each replicate of each tissue. NOTE: if a gene is both “short” and on the MT DNA, it has to be added to the MT count).

Tissue	Column	Total number of short RNA genes	Total number of mt genes	Total number of reads	Total number of reads on short RNAs	% of reads on short RNAs	Total number of reads on mt genes	% of reads on mt genes
Liver	5	7359	15	84897426	73718	16512572	0.19450027	0.0008683184
Liver	6	7359	15	70704038	117116	6741313	0.09534552	0.0016564259
Liver	7	7359	15	70156194	72179	4903999	0.06990116	0.0010288329
Brain	2	7359	15	178485738	36042	1810929	0.01014607	0.0002019321
Brain	3	7359	15	162205306	215583	18885940	0.11643232	0.0013290749
Brain	4	7359	15	146394126	77925	7726701	0.05278013	0.0005322959

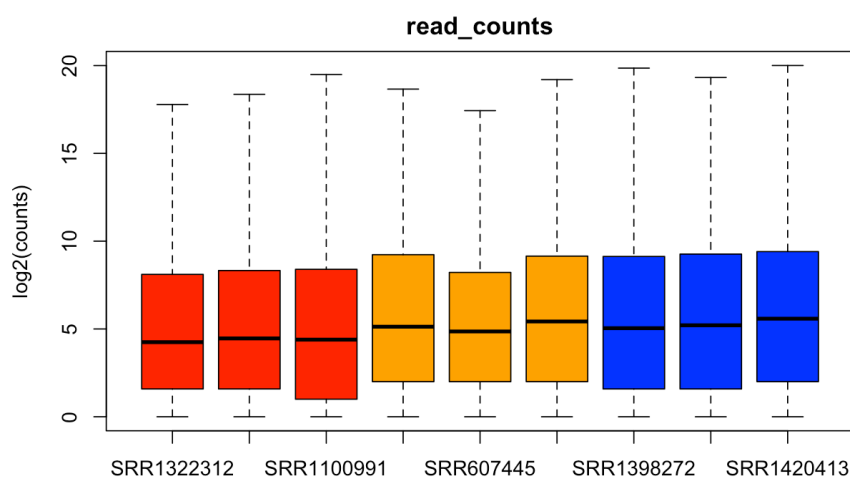
Colon	8	7359	15	75133844	38849	1044689	0.01390437	0.0005170639
Colon	9	7359	15	73981128	58056	4148166	0.05607060	0.0007847407
Colon	10	7359	15	73563982	149686	1043438	0.01418409	0.0020347729

Question 2:

Filtering for genes with low expression and after normalization of the counts according to the default method of the tool you are using (e.g. TMM for edgeR). Then report:

- The total number of genes contained in the initial count table: 58037
- The total number of genes removed because “short RNA” genes (mitochondrial or not): 7359
- The total number of genes removed because on mitochondrial DNA: 15
- The total number of genes removed because of low expression: 23036
- The total number of genes remaining in the count table after the previous three filtering steps: 33834
(the numbers should be the same for all the samples..!)

Insert here a figure with the boxplots of the $\log_2(\text{normalized_counts})$ of each sample, one boxplot per sample. Do not include outliers in the boxplots. Don't forget to label the axes, and write a short caption for the figure.



Liver is red, colon is orange, colon is blue. For the y axis I've used the $\log_2$ of the counts, to make it more easily viewable.

Question 3:

Among the different quality parameters associated with the samples found in the “colData” of the RangedSummarizedExperiment structure containing the data, there is the “RNA Integrity Number” or “RIN”.

Report the RIN values of your samples in a table, and give a short answer to the following questions:

Liver: 6.7 7.9 7.5

Brain 8.9 5 7.1

Colon 6.3 8 5.9

What is the RIN? What are the minimum/maximum values of this number? What is the minimum RIN for a sample to be considered to be “acceptable”? Do all your samples have a RIN number high enough?

The RNA integrity number (RIN) is an algorithm for assigning integrity values to RNA measurements.

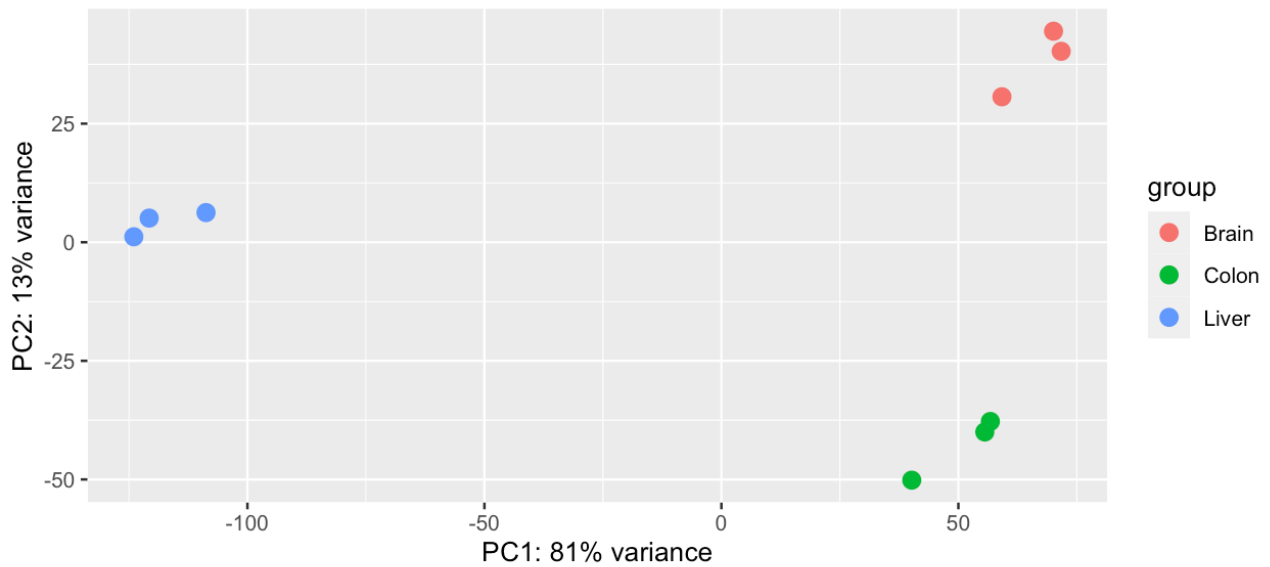
RIN assigns an electropherogram a value of 1 to 10, with 10 being the least degraded.

In the paper below, the threshold is 5.5, so brain3 is the only one below.

The RIN: an RNA integrity number for assigning integrity values to RNA measurements. BMC Molecular Biol 7, 3 (2006)

Question 4:

Exploratory analysis. Using for the expression the log of the normalized counts per million, insert here a figure with the two dimensional spatial representation of your 9 samples. You can use either PCA or MDS as projection technique. Don't forget to label the axes, and write a short caption for the figure.



I used PCA to project them. My samples seem to be well clustered and defined.

Question 5:

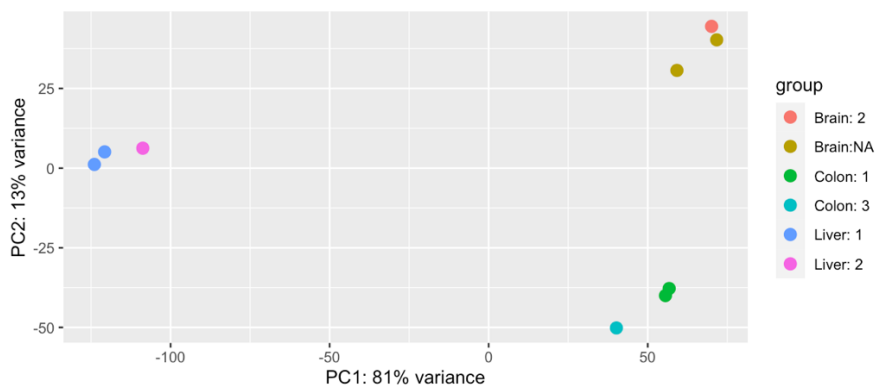
1. a) Look at the figure of the previous question. In your opinion replicates of the same tissue cluster “well” together? If not or in case of doubt, post your figure in the Ariel forum of this course and ask for advice.

I think the clusters are adequately separated and well defined.

2. b) If the replicates cluster “well”, assuming that there are no technical issues introducing a systematic bias in the data, what in your opinion could be the main biological factors making two replicates of the same tissue more or less similar (and the corresponding points more or less close in the two dimensional space)? You might want to look at how samples were collected in the GTEx project to formulate a hypothesis. The reason might be different in the different tissues.

The differences might be explained by the cell type, the age and gender of the donor. Also the low RIN might explain the difference in Brain3, which is slightly off.

3. c) Insert here the plot of Question 4, labelling the samples with one of the factors (or one for each tissue) you considered to be relevant in the previous question.



Smatsscr may explain the differences within the liver and the colon samples (smcenter gives the same result)



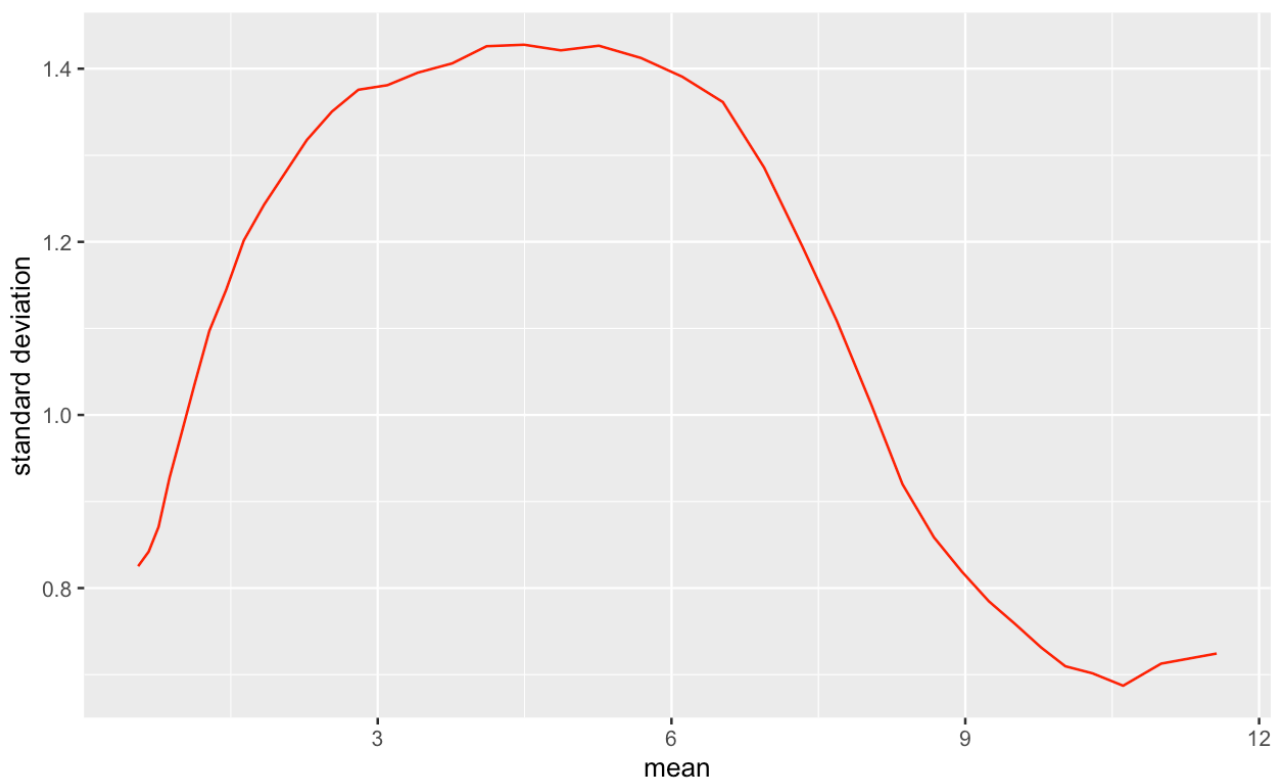
Smnabtcht may explain the differences among the brain samples, but they could also be due to the different neural complexes the three samples come from.

Question 6:

What is the distribution used by the tool you have chosen to model the variability of read counts?

All the tools we have seen permit to generate a plot showing the dependence between mean and variability of expression (using different ways to represent variability, e.g. variance, standard deviation, BCV). Insert here the figure with the mean/variability plot generated by your tool –

explaining what is plotted (which measures are used for the x and y axis), and how the measure is related to the statistical modelling of the tool you employed. Don't forget to label the axes, and write a short caption for the figure.



On the y axis the standard deviation, on the x axis the mean of the counts.

Question 7:

Identify differentially expressed genes (up- and down-regulated) between:

1. a) Tissue 1 and Tissue 2
2. b) Tissue 1 and Tissue 3
3. c) Tissue 2 and Tissue 3

Use the same thresholds for FDR and log2FC in all the comparisons. Then, for each tissue, determine the lists of genes that are:

1. d) Up-regulated in both comparisons against the other two tissues
2. e) Down-regulated in both comparisons against the other two tissues

Prepare an Excel document with the results, one sheet for each of the three comparisons a-b-c and one sheet per tissue for comparisons d-e. Add it to the "Supplementary Material" of your analysis.

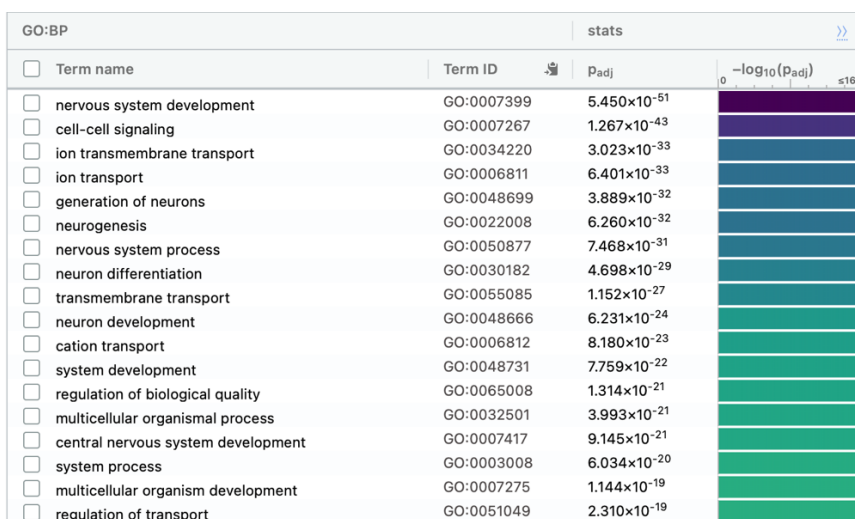
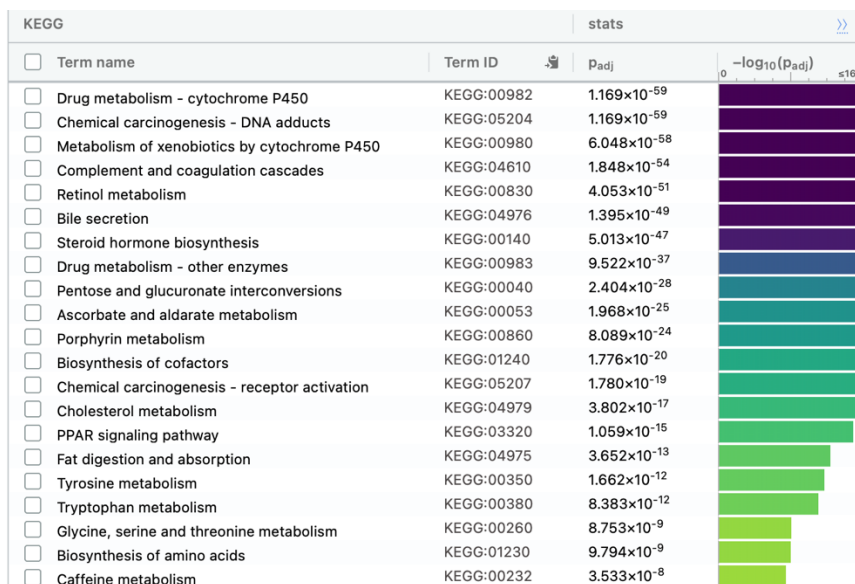
Which thresholds did you employ for the analysis, and why? Just saying "too many" or "too few" genes is not enough – or at least explain why you consider a given number of DE genes to be too high or too low..

I used a logfoldchange of 5 and even so I got far more results than expected.

Question 8:

Functional enrichment/GO/pathway analysis on DE genes. You can use any tool you want. Do you find functional categories associated with genes up-regulated in each tissue with respect to the other two that can be considered “congruent” with the tissue itself and/or the function of its cells? Insert here a figure showing the output(s) of the tool(s) you employed and highlighting the relevant annotation(s) – or showing that you did not find anything relevant. **You can use any functional annotation you want, except those related to “tissue specific expression” of genes.**

Below I have the g:profiler results for liver, brain and colon (setting term size to 1000 max). The first two gave very specific and clear results, while colon not so much. I have showed three different databases (KEGG for liver, BP for brain and CC for colon) as an example.



GO:CC			stats		
☐ Term name	Term ID	P _{adj}	$-\log_{10}(P_{adj})$		
☐ contractile fiber	GO:0043292	6.025×10^{-10}			
☐ myofibril	GO:0030016	5.826×10^{-9}			
☐ sarcomere	GO:0030017	3.055×10^{-8}			
☐ supramolecular fiber	GO:0099512	3.791×10^{-8}			
☐ supramolecular polymer	GO:0099081	4.419×10^{-8}			
☐ Z disc	GO:0030018	1.233×10^{-6}			
☐ I band	GO:0031674	2.872×10^{-6}			
☐ actin cytoskeleton	GO:0015629	8.014×10^{-6}			
☐ actin filament bundle	GO:0032432	3.165×10^{-4}			
☐ stress fiber	GO:0001725	3.755×10^{-3}			
☐ contractile actin filament bundle	GO:0097517	3.755×10^{-3}			
☐ actomyosin	GO:0042641	6.326×10^{-3}			
☐ costamere	GO:0043034	1.376×10^{-2}			
☐ focal adhesion	GO:0005925	2.236×10^{-2}			
☐ cell-substrate junction	GO:0030055	2.580×10^{-2}			
☐ muscle thin filament tropomyosin	GO:0005862	3.003×10^{-2}			
☐ striated muscle thin filament	GO:0005865	3.067×10^{-2}			
☐ actin filament	GO:0005884	3.662×10^{-2}			

Question 9:

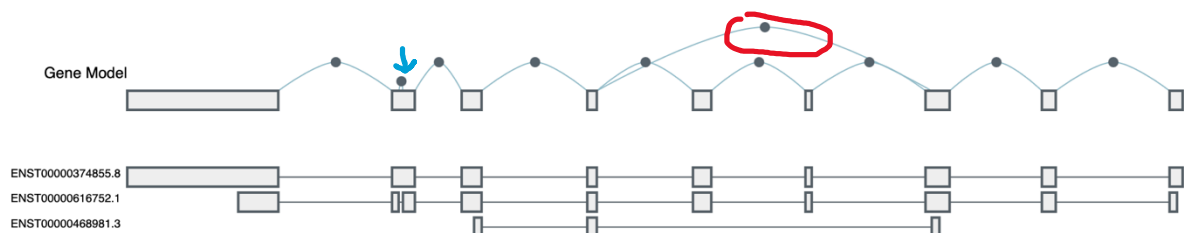
Choose the gene up-regulated in Tissue 1 vs Tissue 2 with a) the lowest FDR b) up-regulated also in Tissue 1 vs Tissue 3 and c) with at least two alternative transcripts according to the GENCODE v25 annotation used by Recount2 (for this point you have to locate and examine it with the UCSC Genome Browser).

ENSG00000136872.17 ALDOB

Insert here a figure showing it in the genome browser itself (the annotation is GENCODE v25 on the latest human assembly).

I found the requested gene at the link below, however I found the gTEX view more easily readable.

http://genome-euro.ucsc.edu/cgi-bin/hgTracks?db=hg38&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr9%3A101421439%2D101449664&hgsid=286312407_Y07tT6b2pdCG9qLwi6iMYvzrBYJg



How many alternative transcripts you can find associated with the gene in the GENCODE annotation used? Does it have alternative promoters? Alternative transcription termination sites? Which modes of alternative splicing can be observed? It might help to mark the different alternative transcription/splicing modes detected on the figure of the gene.

Yes, in liver the median TPM is 2547, while in brain it's around 0.5 and in colon it's below 2. Human protein Atlas seems to confirm these results.